Course 6013B | Semester VI | Theory | 4 Credits

Solid Waste Management & Air Pollution

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Civil Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6013B as Solid Waste Management & Air Pollution in Semester VI for Civil Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Guwahati’s Municipal Solid Waste Management course and NPTEL IIT Roorkee’s Environmental Engineering course. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Landfill designs, incinerator operations, air quality monitoring reports and waste management plans must be based on approved environmental standards, authorised site data, current legislation and competent professional review.

Verified source note: Verified sources support solid waste evolution, sources, types, generation, handling, collection, transport, processing, incineration, composting, anaerobic digestion, landfills, ISWM, air pollution sources, impacts, monitoring (NAAQI) and control equipment. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Solid Waste Management & Air Pollution studies the systematic management of municipal solid waste and the control of atmospheric pollutants. It develops the ability to plan waste collection and transport, evaluate processing and disposal methods (composting, incineration, landfills), monitor air quality, and design air pollution control systems while respecting the technical and regulatory boundaries of environmental protection.

> **Malayalam support:** □ handbook □□□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Environmental studies, basic chemistry, elementary physics, engineering mathematics and environmental awareness.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain solid waste fundamentals, sources, types and the factors affecting waste generation.
2. Explain solid waste handling, collection, transport and processing methods including material recovery.
3. Explain waste disposal technologies, including incineration, composting, anaerobic digestion and sanitary landfills.
4. Explain air pollution sources, impacts, monitoring (NAAQI) and the principles of air pollution control equipment.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□□ exam-□ □□□□□ □□□□□□□□ □□□□□□□ □□□□□ □□□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□□.

Official Course Outcomes		
CO	**Official outcome**	**Bloom level**
CO1	Explain solid waste generation, characteristics and the functional elements of waste management.	Understand
CO2	Explain waste collection systems, transport methods and component separation techniques.	Understand
CO3	Explain waste-to-energy, composting, anaerobic digestion and landfill disposal methods.	Understand
CO4	Explain air pollution sources, monitoring standards and the principles of control equipment.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	Solid Waste Fundamentals and Generation	Evolution of solid waste; sources and types (municipal, industrial, hazardous); sampling and physical/chemical characteristics; estimation of waste quantity; factors affecting generation rate.	Approx. 14 hours	CO1	Module 1
2	Waste Handling, Collection and Transport	Handling and separation at source; primary and secondary collection; collection systems (hauled container, stationary container); transfer stations and transport methods; material recovery facilities (MRF).	Approx. 16 hours	CO2	Module 2
3	Waste Disposal and Energy Recovery	Incineration and waste-to-energy; composting (aerobic, vermicomposting); anaerobic digestion (biogas); sanitary landfills (site selection, leachate/gas management); integrated solid waste management (ISWM).	Approx. 16 hours	CO3	Module 3
4	Air Pollution Monitoring and Control	Air pollution sources and impacts; ambient air quality monitoring (NAAQI); air pollution control equipment (cyclones, ESP, scrubbers); air pollution legislation and environmental acts.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Solid Waste Fundamentals and Generation

Evolution of solid waste; sources and types (municipal, industrial, hazardous); sampling and physical/chemical characteristics; estimation of waste quantity; factors affecting generation rate.

CO1 · Approx. 14 hours

Waste Handling, Collection and Transport

Handling and separation at source; primary and secondary collection; collection systems (hauled container, stationary container); transfer stations and transport methods; material recovery facilities (MRF).

CO2 · Approx. 16 hours

Waste Disposal and Energy Recovery

Incineration and waste-to-energy; composting (aerobic, vermicomposting); anaerobic digestion (biogas); sanitary landfills (site selection, leachate/gas management); integrated solid waste management (ISWM).

CO3 · Approx. 16 hours

Air Pollution Monitoring and Control

Air pollution sources and impacts; ambient air quality monitoring (NAAQI); air pollution control equipment (cyclones, ESP, scrubbers); air pollution legislation and environmental acts.

CO4 · Approx. 14 hours

MODULE 1

Solid Waste Fundamentals and Generation

Evolution of solid waste; sources and types (municipal, industrial, hazardous); sampling and physical/chemical characteristics; estimation of waste quantity; factors affecting generation rate.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

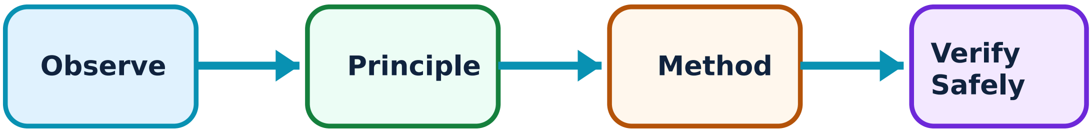
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Solid Waste Fundamentals and Generation: identify the system, state its principle, follow the correct method and verify the result safely.

1. Solid waste types and sources−

Solid waste includes municipal, industrial, agricultural and hazardous wastes. Municipal Solid Waste (MSW) consists of household, commercial and institutional waste. Understanding sources and types is essential for selecting appropriate management strategies.

Study and application method

Classify five types of solid waste by source and describe their typical physical and chemical characteristics.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Classify five types of solid waste by source and describe their typical physical and chemical characteristics.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Hazardous waste requires specialized handling and disposal. Do not mix hazardous waste with municipal solid waste.

2. Waste generation and characterisation−

Waste generation rates depend on population, lifestyle and economic activity. Characterisation involves measuring moisture content, density, organic/inorganic fractions and energy content, which determines the suitability of disposal methods like composting or incineration.

Study and application method

Calculate the moisture content of a waste sample conceptually; list three factors that increase waste generation in urban areas.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Calculate the moisture content of a waste sample conceptually; list three factors that increase waste generation in urban areas.

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Common mistake / precaution: Waste generation data is highly variable. Use current, representative local data for planning waste management infrastructure.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Waste Handling, Collection and Transport

Handling and separation at source; primary and secondary collection; collection systems (hauled container, stationary container); transfer stations and transport methods; material recovery facilities (MRF).

Approx. 16 hours **CO2** **Understand · Apply**

Learning goals

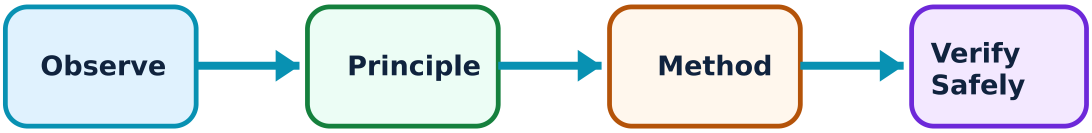
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Waste Handling, Collection and Transport: identify the system, state its principle, follow the correct method and verify the result safely.

1. Collection and transport systems−

Collection is the most expensive part of waste management. Systems include door-to-door collection and community bins. Transfer stations optimize transport by consolidating waste from smaller vehicles into larger ones for long-distance haulage.

Study and application method

Compare a hauled-container system with a stationary-container system; describe the role of a transfer station in urban waste management.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare a hauled-container system with a stationary-container system; describe the role of a transfer station in urban waste management.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Inefficient collection leads to illegal dumping and health hazards. Route optimization and regular vehicle maintenance are critical.

2. Processing and component separation−

Processing at source and at Material Recovery Facilities (MRF) involves separating recyclable components (paper, plastic, metal) from the waste stream using unit operations like screening, magnetic separation and air classification.

Study and application method

Sketch a flow diagram for a material recovery facility (MRF) showing three unit operations for component separation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch a flow diagram for a material recovery facility (MRF) showing three unit operations for component separation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Component separation involves mechanical and safety risks. Use appropriate personal protective equipment (PPE) and follow safety protocols in MRFs.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Waste Disposal and Energy Recovery

Incineration and waste-to-energy; composting (aerobic, vermicomposting); anaerobic digestion (biogas); sanitary landfills (site selection, leachate/gas management); integrated solid waste management (ISWM).

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

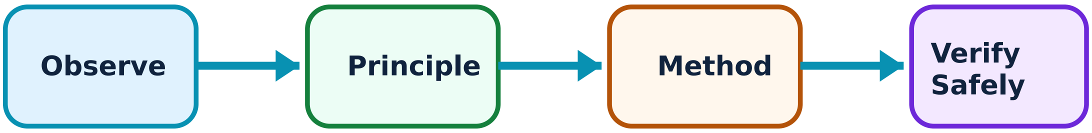
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Waste Disposal and Energy Recovery: identify the system, state its principle, follow the correct method and verify the result safely.

1. Biological and thermal treatment−

Biological methods like composting and anaerobic digestion recover nutrients and energy from organic waste. Thermal methods like incineration reduce waste volume and generate electricity, requiring strict flue gas treatment to prevent air pollution.

Study and application method

Compare aerobic composting and anaerobic digestion based on their products and environmental requirements; list three benefits of waste-to-energy plants.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare aerobic composting and anaerobic digestion based on their products and environmental requirements; list three benefits of waste-to-energy plants.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Incineration produces toxic solid residues and emissions. All plants must meet authorised environmental standards for discharge.

2. Sanitary landfills and ISWM−

Landfilling is the final disposal method for non-recyclable waste. Sanitary landfills use liners and covers to prevent leachate (liquid) and gas (methane) from contaminating the environment. Integrated Solid Waste Management (ISWM) coordinates all functional elements for sustainability.

Study and application method

Draw a cross-section of a sanitary landfill showing the liner, leachate collection system and final cover; explain the hierarchy of ISWM.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw a cross-section of a sanitary landfill showing the liner, leachate collection system and final cover; explain the hierarchy of ISWM.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Uncontrolled dumping (open dumps) leads to groundwater pollution and disease. Do not dispose of waste in unlined or unauthorised sites.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Air Pollution Monitoring and Control

Air pollution sources and impacts; ambient air quality monitoring (NAAQI); air pollution control equipment (cyclones, ESP, scrubbers); air pollution legislation and environmental acts.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

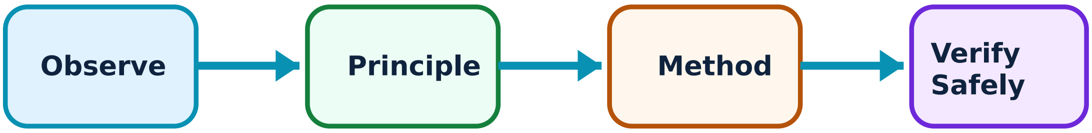
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Air Pollution Monitoring and Control: identify the system, state its principle, follow the correct method and verify the result safely.

1. Air quality monitoring and impacts−

Air pollutants (particulates, SOx, NOx, CO) are monitored using ambient air quality standards (NAAQI). Impacts include respiratory health issues, acid rain and climate change. Monitoring provides the data for regulatory compliance and public health alerts.

Study and application method

Identify the major air pollutants monitored under NAAQI and describe their primary health impacts.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify the major air pollutants monitored under NAAQI and describe their primary health impacts.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Air quality data is sensitive to sensor location and calibration. Use only authorised monitoring data for official health assessments.

2. Air pollution control equipment−

Control equipment is selected based on pollutant type and size. Particulates are removed using cyclones (centrifugal), fabric filters (filtration) and electrostatic precipitators (electrical). Gases are removed using scrubbers (absorption) or adsorption.

Study and application method

Compare the working principles of a fabric filter and a wet scrubber; list three environmental acts in India that regulate air pollution.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare the working principles of a fabric filter and a wet scrubber; list three environmental acts in India that regulate air pollution.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Control equipment must be properly maintained to ensure continuous compliance. Failure of control devices can lead to severe local air quality degradation.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Solid Waste Fundamentals and Generation

Use the concept flow to revise observation, principle, method and verification.

Module 2: Waste Handling, Collection and Transport

Use the concept flow to revise observation, principle, method and verification.

Module 3: Waste Disposal and Energy Recovery

Use the concept flow to revise observation, principle, method and verification.

Module 4: Air Pollution Monitoring and Control

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Classify a supplied list of solid waste items by source and recyclability.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the functional elements of an integrated solid waste management system.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare composting and incineration based on their environmental impact and resource recovery.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw a cross-section of a sanitary landfill and label the environmental protection layers.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify the air pollution control devices suitable for a local thermal power plant or cement factory.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Solid Waste Fundamentals and Generation

Explain Solid waste types and sources and state one correct method, application or precaution.−

Solid waste includes municipal, industrial, agricultural and hazardous wastes. Municipal Solid Waste (MSW) consists of household, commercial and institutional waste. Understanding sources and types is essential for selecting appropriate management strategies.

Answer framework: Classify five types of solid waste by source and describe their typical physical and chemical characteristics.

Important point: Hazardous waste requires specialized handling and disposal. Do not mix hazardous waste with municipal solid waste.

Explain Waste generation and characterisation and state one correct method, application or precaution.−

Waste generation rates depend on population, lifestyle and economic activity. Characterisation involves measuring moisture content, density, organic/inorganic fractions and energy content, which determines the suitability of disposal methods like composting or incineration.

Answer framework: Calculate the moisture content of a waste sample conceptually; list three factors that increase waste generation in urban areas.

Important point: Waste generation data is highly variable. Use current, representative local data for planning waste management infrastructure.

Module 2 — Waste Handling, Collection and Transport

Explain Collection and transport systems and state one correct method, application or precaution.−

Collection is the most expensive part of waste management. Systems include door-to-door collection and community bins. Transfer stations optimize transport by consolidating waste from smaller vehicles into larger ones for long-distance haulage.

Answer framework: Compare a hauled-container system with a stationary-container system; describe the role of a transfer station in urban waste management.

Important point: Inefficient collection leads to illegal dumping and health hazards. Route optimization and regular vehicle maintenance are critical.

Explain Processing and component separation and state one correct method, application or precaution.−

Processing at source and at Material Recovery Facilities (MRF) involves separating recyclable components (paper, plastic, metal) from the waste stream using unit operations like screening, magnetic separation and air classification.

Answer framework: Sketch a flow diagram for a material recovery facility (MRF) showing three unit operations for component separation.

Important point: Component separation involves mechanical and safety risks. Use appropriate personal protective equipment (PPE) and follow safety protocols in MRFs.

Module 3 — Waste Disposal and Energy Recovery

Explain Biological and thermal treatment and state one correct method, application or precaution.−

Biological methods like composting and anaerobic digestion recover nutrients and energy from organic waste. Thermal methods like incineration reduce waste volume and generate electricity, requiring strict flue gas treatment to prevent air pollution.

Answer framework: Compare aerobic composting and anaerobic digestion based on their products and environmental requirements; list three benefits of waste-to-energy plants.

Important point: Incineration produces toxic solid residues and emissions. All plants must meet authorised environmental standards for discharge.

Explain Sanitary landfills and ISWM and state one correct method, application or precaution.−

Landfilling is the final disposal method for non-recyclable waste. Sanitary landfills use liners and covers to prevent leachate (liquid) and gas (methane) from contaminating the environment. Integrated Solid Waste Management (ISWM) coordinates all functional elements for sustainability.

Answer framework: Draw a cross-section of a sanitary landfill showing the liner, leachate collection system and final cover; explain the hierarchy of ISWM.

Important point: Uncontrolled dumping (open dumps) leads to groundwater pollution and disease. Do not dispose of waste in unlined or unauthorised sites.

Module 4 — Air Pollution Monitoring and Control

Explain Air quality monitoring and impacts and state one correct method, application or precaution.−

Air pollutants (particulates, SOx, NOx, CO) are monitored using ambient air quality standards (NAAQI). Impacts include respiratory health issues, acid rain and climate change. Monitoring provides the data for regulatory compliance and public health alerts.

Answer framework: Identify the major air pollutants monitored under NAAQI and describe their primary health impacts.

Important point: Air quality data is sensitive to sensor location and calibration. Use only authorised monitoring data for official health assessments.

Explain Air pollution control equipment and state one correct method, application or precaution.−

Control equipment is selected based on pollutant type and size. Particulates are removed using cyclones (centrifugal), fabric filters (filtration) and electrostatic precipitators (electrical). Gases are removed using scrubbers (absorption) or adsorption.

Answer framework: Compare the working principles of a fabric filter and a wet scrubber; list three environmental acts in India that regulate air pollution.

Important point: Control equipment must be properly maintained to ensure continuous compliance. Failure of control devices can lead to severe local air quality degradation.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Solid Waste Fundamentals and Generation.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.

3. State one key definition from Module 2: Waste Handling, Collection and Transport.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Waste Disposal and Energy Recovery.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Air Pollution Monitoring and Control.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Evolution of solid waste; sources and types (municipal, industrial, hazardous); sampling and physical/chemical characteristics; estimation of waste quantity; factors affecting generation rate..
2. Develop a complete answer or practical plan for: Handling and separation at source; primary and secondary collection; collection systems (hauled container, stationary container); transfer stations and transport methods; material recovery facilities (MRF)..
3. Develop a complete answer or practical plan for: Incineration and waste-to-energy; composting (aerobic, vermicomposting); anaerobic digestion (biogas); sanitary landfills (site selection, leachate/gas management); integrated solid waste management (ISWM)..
4. Develop a complete answer or practical plan for: Air pollution sources and impacts; ambient air quality monitoring (NAAQI); air pollution control equipment (cyclones, ESP, scrubbers); air pollution legislation and environmental acts..

LAST-MILE REVISION

Quick Revision

Module 1: Solid Waste Fundamentals and Generation

Evolution of solid waste; sources and types (municipal, industrial, hazardous); sampling and physical/chemical characteristics; estimation of waste quantity; factors affecting generation rate.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Waste Handling, Collection and Transport

Handling and separation at source; primary and secondary collection; collection systems (hauled container, stationary container); transfer stations and transport methods; material recovery facilities (MRF).

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Waste Disposal and Energy Recovery

Incineration and waste-to-energy; composting (aerobic, vermicomposting); anaerobic digestion (biogas); sanitary landfills (site selection, leachate/gas management); integrated solid waste management (ISWM).

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Air Pollution Monitoring and Control

Air pollution sources and impacts; ambient air quality monitoring (NAAQI); air pollution control equipment (cyclones, ESP, scrubbers); air pollution legislation and environmental acts.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
Solid waste types and sources	Solid waste includes municipal, industrial, agricultural and hazardous wastes.
Waste generation and characterisation	Waste generation rates depend on population, lifestyle and economic activity.
Collection and transport systems	Collection is the most expensive part of waste management.
Processing and component separation	Processing at source and at Material Recovery Facilities (MRF) involves separating recyclable components (paper, plastic, metal) from the waste stream using unit operations like screening, magnetic separation and air classification.
Biological and thermal treatment	Biological methods like composting and anaerobic digestion recover nutrients and energy from organic waste.
Sanitary landfills and ISWM	Landfilling is the final disposal method for non-recyclable waste.
Air quality monitoring and impacts	Air pollutants (particulates, SOx, NOx, CO) are monitored using ambient air quality standards (NAAQI).
Air pollution control equipment	Control equipment is selected based on pollutant type and size.

LEARNING RESOURCES

References

Recommended waste-and-air-pollution references

1. Ajay Kalamdhad, Municipal Solid Waste Management.
2. George Tchobanoglous, Integrated Solid Waste Management.
3. M. N. Rao and H. V. N. Rao, Air Pollution.
4. S. K. Garg, Environmental Engineering (Vol. II).

Approved public waste-and-air-pollution sources

- https://nptel.ac.in/courses/105103205
- https://onlinecourses.nptel.ac.in/noc23_ce66/preview

These sources provide the verified study basis while official Course 6013B detail is unavailable; they do not replace official environmental compliance reports, authorised landfill designs, official air-quality data or legal environmental mandates.